Aneesh Grover

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Education

University of Toledo **Toledo, OH**
Bachelor of Science in Computer Science and Engineering

Relevant Coursework

- Data Structures and Algorithms
- Operating Systems
- Discrete Structures
- Object-Oriented Programming
- Computer Architecture
- Artificial Intelligence

Experience

AcademicAlly (ACM Thapar Society Project) **July 2024 – August 2025**
Software Developer
- Developed and maintained **AcademicAlly**, a responsive timetable portal serving **10,000+ students** for fast, reliable class schedule lookup.
- Converted **50,000+ Excel rows** into optimized JSON and engineered backend using Python, Pandas, and Openpyxl, implementing **custom error-handling** to reduce parsing errors by **95%**.
- Deployed and maintained the **live portal** with automated updates, bug tracking, and continuous improvements, ensuring seamless user experience.

Projects

PatchIt - Civic Infrastructure Monitoring
- Co-led the development of a **crowdsourced platform** connecting citizens with contractors to monitor and repair civic infrastructure, improving transparency and efficiency in road maintenance.
- Designed a **YOLO-based pothole detection pipeline**, achieving **85.8% precision** and **81.7% mAP@50** on real-world datasets, and integrated **Depth Anything V2 to estimate 3D geometry** and approximate repair costs.
- Recognized as **Runner-up (Open Innovation)** at **NVIDIA's SatHack Hackathon.**

Indoor Navigation System - Proof of Concept
- Developed a full-stack indoor navigation platform **(Next.js, FastAPI, Supabase)** with real-time floorplan editing and dynamic route rendering.
- Implemented version-controlled JSON storage to track changes to the floorplan and facilitate easy rollback.
- Deployed on Vercel; exploring integration with large retail spaces for in-store guidance.

DeepResearch - Multi-Agent Research Assistant
- Built a LangGraph-based multi-agent pipeline integrating **Tavily API** and **Groq LLaMA3** for automated research synthesis.
- Implemented **FAISS-based retrieval** with HuggingFace embeddings for contextual content indexing
- Citation-aware RAG workflow to ensure responses are traceable and backed by reliable sources.

Audio Deepfake Detection
- Researched and selected lightweight, state-of-the-art audio deepfake detection architectures for **mobile and edge deployment**.
- Enhanced the codebase by resolving technical bottlenecks, adding a **dedicated inference pipeline**, and introducing **checkpointing** for **stable model training**.
- Delivered highly accurate results under limited GPU resources, achieving **3.37% Equal Error Rate** and **0.10 t-DCF**, demonstrating potential for **efficient on-device deployment**.

Skills

Programming: Python, Java, C++, JavaScript, TypeScript
Frameworks & Libraries: Next.js, FastAPI, PyTorch, TensorFlow, transformers, OpenCV
Software Engineering: OOP, system design, REST APIs, debugging, testing, version control, CI/CD
Machine Learning & Vision: Object detection, depth estimation, CUDA, FAISS, RAG
Tools & Platforms: Git, Supabase, Streamlit, Manim, Google APIs

Leadership & Involvement

Association for Computing Machinery (ACM) Thapar **Nov 2023 – Aug 2025**
Executive Board (Community Collaboration) & Machine Learning Bootcamp Lead
- Organized ACM Thapar's flagship hackathon Hacklipse 2024, hosting 300+ participants.
- Mentored 100+ students through ACM's ML Bootcamp, designing curriculum and weekly sessions.

Achievements

- Secured **runner-up** in **Open Innovation Category** at **NVIDIA's SatHack**, competing against **1500+ participants**.
- Awarded Best Use Case at ACM Projectathon 2024 for AcademicAlly